

games and apps. There are many companies in the segment that are offering powerful and faster processors to meet consumer demands. Recently, a processor was launched with features like real-time processing performance with extended connectivity and intuitive user interface (UI), and it could perform tasks such as instantly downloading and running apps. For example, Tegra x1 processor from NVIDIA caters the requirement for PC gaming and mobile devices.

**Servers.** In servers, a new graphic processing unit (GPU) built for high-performance computing has been launched. The server GPU is supported by a powerful software ecosystem, helping developers to better harness its compute performance including support for OpenCL 2.0. The device also provides support to handle workloads in a variety of sectors by enabling OpenMP, an API for high-level parallelism in C, C++ and FORTRAN languages. An example would be AMD FirePro S9150, which offers these solutions for the servers.

**Digital signages.** A set back box (SBB) is largely used by digital media players. SBB is basically a consumer device, or box, which enables the user to access both linear broadcast and Internet based content, with a range of other services like an electronic programme guide (EPG), pay per view (PPV) and video on-demand (VOD), enabling the user to view over large-screen sets.

The arrival of the new accelerated processing units (APUs) will turn the conventional digital signage displays into a wide range of business needs. These APUs will help media players to deliver breakthrough graphics performance and support multi-video streaming up to two displays, all being achieved in a power-efficient and ultra-power-compact form factor. For instance, embedded RX425BB APU from AMD is offering a solution in the digital signage medium.

**Low-power chipsets.** Low-power chipset is the next big trendsetter for the power-hungry video and audio an-

## MPEG-H audio

In recent times, multimedia technology in car audio systems, TV shows, movies, live sports is getting more and more interactive for a better sound experience. Recently, an audio element was launched where the audio component came with MPEG-H audio, a technology designed to help content creators craft digital 3D soundscapes with greater depth and realism. Advantages of MPEG-H audio are engineered to go beyond 5.1 or even 7.1 surround sound, to paint a more vivid soundscape than has ever been possible before. What sets MPEG-H audio apart from others is its support for the new scene based audio representation (also known as higher order ambisonics, or HOA), in addition to traditional channel and object based formats.

Scene based audio can be used for both live capture and recorded content, and can fit into existing infrastructure for audio broadcast and streaming. MPEG-H allows scene based audio to be transmitted simultaneously with other audio objects. This means, people watching a sporting event could choose commentary in their preferred language, select a preferred commentator or mute commentary altogether. Scene based audio also lets the end user focus on a specific direction in the sound field and rotate the same. A company like Qualcomm Technologies is among the players who are revolutionising multimedia player technology.

alytics. A new DSP has been launched that delivers up to 200MHz performance, enabling faster processing of data with low power consumption for demanding applications, such as audio and video, biometrics and other analytic-specific applications. These applications are opening up a world of possibilities that add capabilities such as face detection, object tracking and voice recognition to a new generation of smart products. These low-power chipsets offer a combination of low standby and low active power, making these the best choice for battery-powered analytic portable systems. Products like Texas Instruments (TI) TMS320C5517 chipset are offering solutions in this domain.

**Digital cockpits.** Head-up displays (HUDs) and advanced driver assistance systems (ADAS) are going to be the next game changers in the automotive industry. On an HUD, you can project information on the windshield and inform the driver where to turn, notify lane markings and notify how close the car is actually to the car in front, while features like traffic-sign recognition, blind-spot detection, driver-alertness monitoring and night vision will be defined under ADAS.

Chipsets have been launched recently serving the need of this specific application, which combine the capabilities of projecting information on a surface providing automotive-

scale reliability and a wide field-of-view. Examples of such chips are TI's DLP3000-Q1, DRA75x Jacinto 6 Ep and Jacinto 6 Ex processors.

**TFT LCDs.** With the emerging demand and application of HUDs, new and advanced TFT LCDs have been recently launched in the market to meet the requirements of the automobile industry. These TFTs are capable of displaying HUD applications and driver information. Some companies like Kyocera Industrial Ceramic have introduced new advanced TFT LCDs to meet the needs of the automobile market.

**Ultra-4k chipsets.** The fast-growing segment in the consumer electronic space is HD viewing. Viewing has now become four times better than its predecessors, through what is known as ultra high definition (UHD), or 4K. Features like active pull-ups on all display data channel (DDC) lines to maintain UHD signal integrity minimise dependence on HDMI cables for better screening experience.

On the other hand, increased bandwidth of 10GHz to pass through 4K signals without distortion, low-clamping voltage of 10V and fast response time increases performance. For instance, STMicroelectronics is one of those companies that have launched single-chip HDMI (HDMI2C1-14HD, HDMI2C1-14HDS,